

E-Commerce Sales Dashboard Analysis

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Executive Summary

This project analyzes an e-commerce sales dataset to identify trends in revenue, profitability, customer behavior, and product performance. Using Microsoft Excel, the dataset was cleaned, validated, and transformed into an interactive dashboard using Pivot Tables, Pivot Charts, KPI cards, and slicers.

The project demonstrates the complete analytics workflow from raw data to business insights.

1. Introduction

Businesses generate large volumes of transactional data every day. Without proper analysis, it is difficult to identify sales trends, customer behavior, and profitable products.

The objective of this project was to transform raw e-commerce data into an interactive dashboard that supports data-driven decision-making.

2. Project Objectives

- Clean and validate the dataset.
 - Perform feature engineering.
 - Build Pivot Tables.
 - Create KPIs.
 - Design an interactive dashboard.
 - Generate business insights.
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3. Dataset Overview

Source:

Kaggle – E-Commerce Orders Dataset

Dataset Size:

- 30,000 Orders
- 41 Columns

Data includes:

- Customer Information
 - Product Information
 - Sales
 - Revenue
 - Profit
 - Payment Methods
 - Membership Status
 - Ratings
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4. Data Cleaning

Completed tasks:

- Duplicate Check
- Missing Value Check
- Data Type Validation
- Date Formatting
- Feature Engineering

Created:

- Month
 - Year
 - Quarter
 - Age Group
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Data Validation

During the cleaning process, I identified an inconsistency in the dataset's profit calculation.

The original workbook calculated:

```
Profit_Amount = Order_Amount × Profit_Margin_Percent
```

Since `Profit_Margin_Percent` was stored as whole-number values (for example, 15.52 instead of 15.52%), the resulting profit values were unrealistically high.

The calculation was corrected to:

```
Profit_Amount = Order_Amount × (Profit_Margin_Percent / 100)
```

After the correction:

- Total Revenue = **\$11.37M**
- Total Profit = **\$2.50M**

This produced financially realistic results for analysis.

5. Dashboard Development

The dashboard contains:

- KPI Cards
 - Pivot Charts
 - Interactive Slicers
 - Revenue Analysis
 - Profit Analysis
 - Customer Analysis
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6. Key Findings

- Bags generated the highest revenue.
 - QuickCart was the top-performing brand.
 - Sales peaked toward the end of the year.
 - Customers aged 36–45 contributed the highest revenue.
 - Standard membership generated the largest share of sales.
 - Revenue reached approximately **\$11.37M** while corrected profit totaled **\$2.50M**.
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7. Business Recommendations

- Focus inventory planning on high-performing product categories.
- Strengthen partnerships with leading brands.

- Develop loyalty campaigns to increase membership upgrades.
 - Expand marketing initiatives targeting customers aged 36–45.
 - Investigate seasonal demand to optimize promotions and inventory planning.
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8. Conclusion

This project demonstrates a complete end-to-end data analytics workflow using spreadsheet tools. From data cleaning and validation to dashboard development and business recommendations, the project highlights the importance of ensuring data quality before conducting analysis.

The final dashboard provides an interactive view of sales performance and supports informed business decisions through clear visualizations and key performance indicators.